

The cellular mechanism of the higher thermal sensitivity of methanogenesis relative to respiration

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Abstract

The thermal sensitivity of methane production is higher than that of respiration, and remarkably this holds consistently from individual methanogen cultures to whole ecosystems¹ — implying the pattern originates at the cellular scale. Yet the cellular mechanism is unknown, and more generally it is unresolved whether the activation energy (E_a) of a metabolic rate reflects one rate-limiting enzyme, an aggregate of the proteome, or a biophysical constant. We address this with mechanistic enzyme- and temperature-constrained genome-scale models (etc-GEMs) — the first for their taxa — of a respiring bacterium (*Escherichia coli*) and a hydrogenotrophic methanogen (*Methanococcus maripaludis*), each calibrated to a measured growth thermal performance curve. Quantifying E_a with the window-independent Sharpe–Schoolfield E , both models reproduce the observed value (*E. coli* 0.68 eV vs observed 0.56, on the “universal” ~0.65 eV benchmark; *M. maripaludis* 1.06 vs 1.04), and the two calibration posteriors are cleanly separated — reproducing the empirical pattern at the cellular scale. A non-circular, control-weighted decomposition shows the +0.38 eV difference is **structural, not thermodynamic**: the methanogen’s enzymes are on average *less* temperature-sensitive (0.51 vs 0.87 eV), but its growth control concentrates on the high- E_a methanogenesis backbone (+0.34 vs +0.11 eV), and its constant-ribosome allocation strategy lacks the proteome buffer that lowers *E. coli*’s E_a (+0.00 vs –0.13 eV). Both terms are grounded in each organism’s own measured allocation physiology. The higher thermal sensitivity of methanogenesis is thus a property of metabolic architecture and resource allocation — the same control-weighted aggregation acting on a narrow, unbuffered energy backbone — providing a concrete cellular origin for a pattern that propagates across scales.

1 Introduction

The temperature dependence of biological rates is one of the most general regularities in biology. Across taxa and levels of organisation, metabolic rates increase with temperature in an approximately Boltzmann–Arrhenius fashion, and the fitted **activation energy** E_a — the slope of $\ln(\text{rate})$ against inverse temperature over the rising limb — clusters near a “universal” value of ~0.6–0.7 eV^{2,3}. This apparent constancy underpins metabolic scaling theory and is widely used to predict how biological processes respond to warming. It also raises an unresolved mechanistic question: is an organism’s E_a set by a single rate-limiting enzyme whose temperature response the organism inherits, by an aggregate over many enzymes, or by a shared thermodynamic constraint? The distinction matters, because only a mechanistic account can explain the systematic *departures* from the universal value that are also observed.

The most consequential such departure concerns methane.¹ showed empirically that the thermal sensitivity of methane flux is substantially higher than that of respiration (and of photosynthesis), and — critically — that this ordering is **consistent from individual methanogen cultures through microbial communities to whole ecosystems**. Cross-scale consistency of an activation energy is a strong clue: it implies that the pattern is not an emergent property of ecological aggregation but originates at the **cellular and enzymatic scale** and propagates upward. What that cellular mechanism is, however, has remained unknown. Why should the growth of a methanogen be more thermally sensitive than that of a respiring bacterium — is it a more temperature-sensitive enzymatic machinery, or something about how that machinery is organised and deployed?

Answering this requires a model in which the organism-level E_a can be traced back to individual enzymes. Enzyme- and temperature-constrained genome-scale models (etc-GEMs) provide exactly this: they impose a temperature-dependent turnover on every enzyme and a shared proteome budget, and predict the whole growth thermal performance curve (TPC) from enzyme-constrained flux balance analysis. Building on a yeast thermal-etcGEM framework⁴, we construct etc-GEMs — to our knowledge the first for either taxon — of *E. coli* (aerobic respiration, a complete electron transport chain) and *M. maripaludis* (hydrogenotrophic methanogenesis, whose *only* energy-conserving pathway is the Wolfe-cycle backbone terminating in the notoriously slow methyl-coenzyme-M reductase, Mcr). We calibrate each to a measured growth TPC, quantify E_a with a window-independent metric, and decompose it non-circularly into a control-weighted aggregate of the per-enzyme activation energies plus named physiological couplings. The answer, previewed here, is that the higher thermal sensitivity of methanogenesis is **structural**: not a more temperature-sensitive enzyme set, but the concentration of growth control on a single high-sensitivity energy backbone together with the absence of the proteome-allocation buffer that dampens the respiring bacterium’s E_a .

2 Results

2.1 Two etc-GEMs reproduce two separated activation energies

Each etc-GEM was calibrated by Bayesian inference to its organism’s measured growth TPC — *E. coli* on rich Brain–Heart–Infusion broth over 7–46 °C⁵, and *M. maripaludis* on H₂/CO₂ defined medium over 15–52 °C⁶. Both reproduce their data: the posterior-predictive median and 90% band pass through the observed points (Fig. 1, left and centre), and the two curves differ visibly in the steepness of their rising limbs. Propagating each calibration posterior through the Sharpe–Schoolfield fitter (Methods) yields the posterior distribution of E_a for each organism. These are **cleanly separated** — *M. maripaludis* at ~1.0 eV and *E. coli* at ~0.6 eV, with non-overlapping 90% credible intervals ([0.88, 1.16] versus [0.60, 0.80]; Fig. 1, right). The empirical pattern of¹ — higher thermal sensitivity for methanogenesis than for respiration — is thus reproduced at the cellular scale by two independent mechanistic models, and is robust to Bayesian parameter uncertainty rather than an artefact of a point estimate. The remainder of the paper asks *why*.

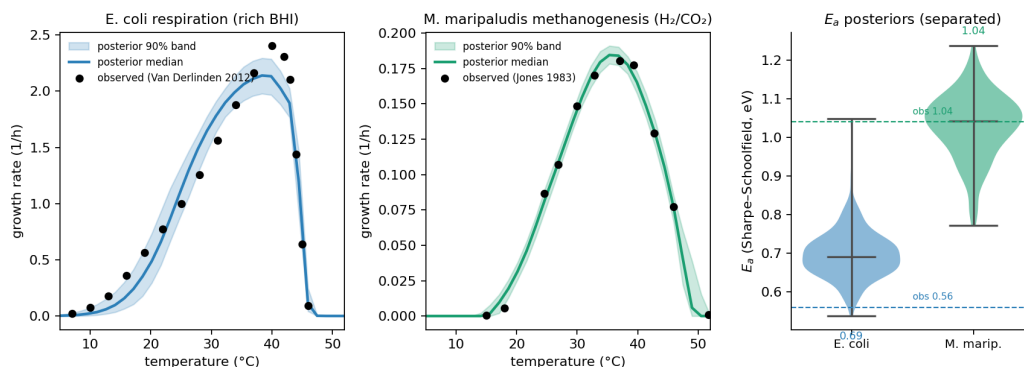


Fig. 1 | Two organisms, two activation energies, reproduced mechanistically. Calibrated (Bayesian) growth thermal performance curves for *E. coli* respiration (rich BHI) and *M. maripaludis* methanogenesis (H₂/CO₂): posterior-predictive median and 90% band with the observed data^{5,6}. Right: the Sharpe–Schoolfield E_a posteriors, propagated from the calibration draws, are separated (methanogen ~1.0 vs *E. coli* ~0.6 eV; dashed lines, observed values).

2.2 Both models predict the observed activation energy

Before decomposing the difference we establish that the modelled E_a is a genuine prediction. This test requires a stable metric, because the naive Boltzmann–Arrhenius slope is not one: the rising limb is curved (enzyme kinetics are not log-linear in inverse temperature), so the fitted slope depends on the arbitrary temperature window. Recomputed under five window conventions, the slope E_a of these very curves spans **0.42–1.41**

eV per curve — larger than the cross-organism differences we seek to explain, and enough that under one window the two organisms appear comparable while under another they are far apart (Table 1). We therefore quantify E_a throughout as the window-independent **Sharpe–Schoolfield E^7** , fit to the entire curve shape (Methods); the windowed slope is retained only as a deprecated cross-check.

Under this metric the models agree with the data. The *E. coli* model gives $E_a = 0.68$ eV against an observed 0.56 eV, sitting on the “universal” ~ 0.65 eV metabolic benchmark; the *M. maripaludis* model gives 1.06 eV against an observed 1.04 (Table 2). Because the per-enzyme activation energies are model *inputs* (below), this agreement is a prediction of the organism-level value, not a fitted output — and the methanogen genuinely sits ~ 1.6 -fold above the respiring bacterium.

Table 1: The naive Arrhenius-slope E_a is window-dependent, the Sharpe-Schoolfield E is not. Slope E_a (eV) for the model and observed curves under five window conventions (range shown), versus the window-independent Sharpe-Schoolfield E . The per-curve slope spread (0.42–1.41 eV) exceeds the cross-organism differences; the Sharpe-Schoolfield E is stable and makes model and observed agree.

curve	slope (10–95%)	slope range	Sharpe–Schoolfield E
<i>E. coli</i> model	0.96	0.76–1.56	0.68
<i>E. coli</i> observed	0.64	0.55–0.96	0.56
methanogen model	1.17	0.89–1.78	1.02
methanogen observed	0.66	0.66–2.07	1.04

2.3 The difference is structural: control weighting and allocation, not enzyme thermodynamics

To find what sets each E_a we decompose it, by metabolic control analysis, into a control-weighted aggregate of the per-enzyme activation energies and four named physiological couplings (Methods, Equation 12). The framing is deliberately non-circular: the per-enzyme activation energies E_i are grounded model inputs, so we do not ask “what is E_a ” but how the organism-level value aggregates them and departs from their naive mean. Fig. 2 and Table 2 give the result for both organisms on the same terms.

The central finding is counter-intuitive. The methanogen’s enzymes are, on average, *less* temperature-sensitive than the respiring bacterium’s: the unweighted mean of the per-enzyme activation energies is 0.51 eV in *M. maripaludis* versus 0.87 eV in *E. coli*. Intrinsic enzyme thermodynamics alone would therefore predict the *opposite* ordering to the one observed. The higher methanogen E_a arises instead from how growth control is distributed and how the proteome is allocated.

First, growth control concentrates on the high- E_a methanogenesis backbone. In a hydrogenotroph, energy metabolism *is* growth: essentially all catabolism flows through the Wolfe cycle, so a small set of energy enzymes — formylmethanofuran dehydrogenase (Fwd), the methyltransferase (Mtr), Mcr, the methylene-tetrahydromethanopterin reductase (Mer), the heterodisulfide reductase (Hdr) and the hydrogenases — carries **89% of the control term**, and these enzymes are themselves high- E_a (mean 0.77 eV against a proteome mean of 0.51). This concentration lifts the effective enzyme activation energy by **+0.34 eV**. In *E. coli*, by contrast, growth control is spread across a broad biosynthetic proteome, and the analogous control-weighting term adds only **+0.11 eV**.

Second, the methanogen lacks the proteome-allocation buffer that lowers *E. coli*’s E_a . In *E. coli*, the Scott-type growth law reallocates proteome toward ribosomes as growth accelerates, which shrinks the metabolic sector at higher temperature and pulls the organism E_a down by **−0.13 eV**. *M. maripaludis*, which maintains a **constant ribosomal proteome fraction** and full methanogenesis capacity across growth rate⁸, applies no such buffer: its allocation term, measured on its own sectorized model, is **+0.00 eV**. Both allocation terms are therefore grounded in each organism’s own measured allocation strategy, making the comparison symmetric rather than an artefact of model structure.

Together these two structural terms — the control-weighting difference (+0.24 eV) and the allocation asymmetry (+0.13 eV) — account for essentially the entire +0.38 eV gap in organism E_a . The two remaining

terms nearly cancel: the methanogen’s lower average enzyme activation energy (-0.36 eV) is offset by a larger **aggregation** term ($+0.39$ eV), the tendency of its narrow, synchronised control on a few backbone enzymes to steepen the composite growth curve where *E. coli*’s broad control flattens it. We note that although the aggregation term is the largest single contribution, it is also the most sensitive to the details of the metric; the robust conclusion — that the difference is set by control weighting and allocation — does not depend on it.

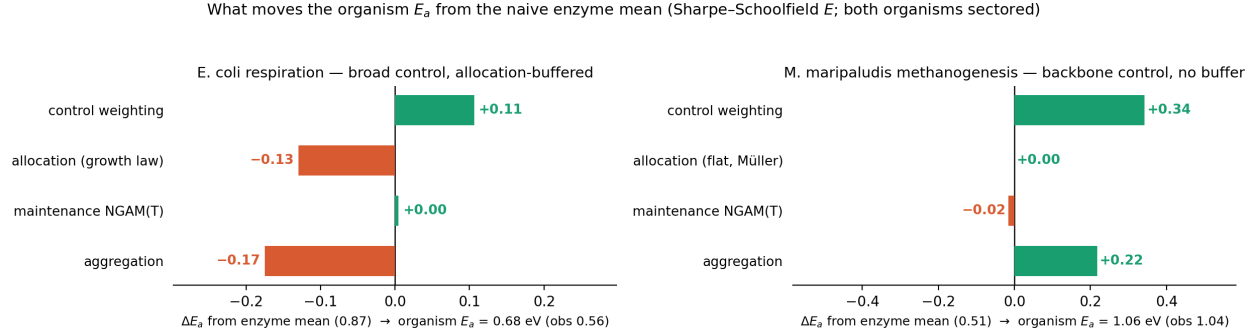


Fig. 2 | What moves the organism E_a away from the naive enzyme mean. Signed contribution of each named term (Sharpe-Schoolfield E ; both organisms carrying a proteome-sector layer), sign-coloured. The higher methanogen E_a is driven by control weighting ($+0.11$ vs $+0.34$ eV) and the allocation asymmetry (-0.13 vs $+0.00$ eV), not by higher intrinsic enzyme temperature-sensitivity (naive mean 0.87 vs 0.51 eV).

Table 2: Symmetric cross-organism decomposition of the Sharpe-Schoolfield activation energy. Both organisms carry a proteome-sector/growth-law layer, so the allocation term is measured for each (*E. coli* Scott growth law; *M. maripaludis* constant-ribosome flat law). The $+0.38$ eV difference is set by control-weighting and allocation, not by a higher intrinsic enzyme E_a (naive mean 0.87 vs 0.51). Both models match the observed SS- E .

term	<i>E. coli</i> (rich BHI)	<i>M. maripaludis</i> (H2/CO2)
organism SS- E E_a (eV)	0.6784	1.0558
observed SS- E (eV)	0.558	1.042
model — observed	0.12	0.014
Topt (°C)	38.4	35.3
CTmax (°C)	46.0	48.8
naive (unweighted) enzyme- E mean	0.8716	0.5118
+ control weighting	0.1063	0.3427
+ allocation (measured; <i>E. coli</i> Scott / methanogen flat)	-0.1295	0
+ maintenance NGAM(T)	0.0049	-0.0164
+ aggregation (heterogeneity/flattening)	-0.1749	0.2177
backbone fraction of control term	n/a (broad)	0.887
sum C_i	0.556	1.189

2.4 The ordering is robust to the uncertain Mcr turnover

The turnover number of the terminal methanogenesis enzyme, Mcr, is uncertain for a mesophile — the literature spans roughly $3\text{--}294\text{ s}^{-1}$. Because Mcr is a candidate rate-controller, we swept its k_{cat} across this range and repeated the analysis (Table 3). The organism E_a remains between 1.01 and 1.27 eV throughout, always well above the *E. coli* value of 0.68, and the methanogenesis backbone carries 83–103% of the control term across the whole range. The ordering and the backbone-control finding are therefore robust to this uncertainty. What does change is the identity of the single leading enzyme: Mcr dominates when it is slow (a low turnover makes it expensive in enzyme mass and thus high in control), Fwd leads when Mcr is fast, and

the two co-lead at the calibrated value. The classic expectation that Mcr specifically sets the methanogen E_a therefore holds only when Mcr is slow; the robust statement is that the *energy backbone as a whole*, not any single enzyme, carries the high thermal sensitivity.

Table 3: Sensitivity to the Mcr turnover number (3-294/s). The organism SS-E stays well above *E. coli*’s 0.68 and the backbone fraction of the control term stays near-total across the range; only the leading enzyme shifts (Mcr when slow, Fwd when fast).

Mcr kcat (1/s)	organism SS-E (eV)	backbone fraction of control	leading enzymes
3	1.2743	1.026	Mcr(1.605);Fwd(0.092);CODH(-0.065)
10	1.0105	0.986	Mcr(0.876);Fwd(0.167);CODH(-0.039)
30	1.0651	0.925	Mcr(0.488);Fwd(0.277);Mtr(0.059)
58.9	1.0558	0.887	Fwd(0.337);Mcr(0.304);Mtr(0.070)
100	1.0659	0.857	Fwd(0.395);Mcr(0.211);Mtr(0.079)
294	1.0171	0.833	Fwd(0.427);Mtr(0.086);Mer(0.079)

3 Discussion

The higher thermal sensitivity of methanogenesis relative to respiration¹ has a concrete cellular mechanism, and it is not the intuitive one. The methanogen’s enzymes are not individually more temperature-sensitive — on average they are less so. Instead the organism-level activation energy is a **control-weighted, allocation-modulated aggregate** of the per-enzyme activation energies, and two structural features of methanogenic physiology raise it. Because a hydrogenotroph conserves energy through a single narrow pathway, growth control concentrates almost entirely on that high-activation-energy energy backbone, rather than being diluted across a broad biosynthetic proteome as in a respiring heterotroph. And because *M. maripaludis* maintains a constant ribosomal fraction and full methanogenesis capacity across growth rate⁸, it lacks the growth-law proteome reallocation that buffers *E. coli*’s activation energy downward. Each term is grounded in the organism’s own measured allocation strategy, so the comparison reflects biology rather than a modelling choice.

This reframes the “universal” metabolic activation energy. Its approximate constancy need not imply a conserved enzyme or a fixed biophysical constant; the same control-weighted-aggregation mechanism, acting on a **different controlling set** and modulated by a **different allocation strategy**, produces systematically different organism-level values. A respiring bacterium lands near the benchmark because its control is broad and allocation-buffered; a methanogen sits well above it because its control is narrow and unbuffered. The corollary is a link, to our knowledge not previously drawn, between **proteome-allocation strategy and thermal sensitivity**: how an organism partitions and couples its proteome is a determinant of how steeply its growth responds to temperature. This connection is significant precisely because of the cross-scale consistency that motivated the question — if the cellular mechanism is control concentration on a high- E_a , unbuffered energy pathway, then that mechanism is what propagates the methanogenesis signal from cultures up to the ecosystem fluxes of¹.

Several caveats bound these conclusions. The aggregation term, though the largest single contribution to the cross-organism difference, is the most sensitive to the definition of the activation-energy metric; we have therefore rested the mechanistic conclusion on the control-weighting and allocation terms, which alone account for the observed gap. The mesophilic turnover of Mcr is the dominant single-parameter uncertainty: the ordering and the backbone-control result are robust across its plausible range, but the identity of the single most rate-controlling enzyme is not, and a direct measurement of mesophilic Mcr k_{cat} would sharpen the attribution. The per-enzyme activation energies are grounded inputs (from macromolecular rate theory, sequence-based optima, a melting proteome, and — for the methanogen — a deep-learning turnover predictor corrected against literature kinetics), so the result is an attribution given those inputs rather than an ab-initio

prediction of the numerical value. Finally, this is a two-organism comparison; the mechanism is a hypothesis that a broader survey should test.

The natural next test is the third process in the¹ ordering. Photosynthesis has an even lower thermal sensitivity than respiration, and a cyanobacterial etc-GEM — with its light-driven energy metabolism and distinct proteome economy — would provide a demanding, directional test of whether the control-and-allocation framework generalises across all three metabolic strategies. If it does, a phenomenological “universal” activation energy will have been replaced by a mechanistic, organism-specific, and predictive account of thermal sensitivity.

4 Methods

We represent each organism by the same layered enzyme- and temperature-constrained genome-scale model (etc-GEM), and write out the full constraint system below before describing where the two taxa depart. The thermal core follows the yeast etc-GEM of⁴; we take from existing work the GECKO/sMOMENT enzyme constraint^{9,10}, macromolecular rate theory¹¹, and the proteome-sector growth law^{12,13}, and add: application to two new taxa; measured- and machine-learning-grounded per-enzyme parameters; an **organism-specific** allocation law; an exact-likelihood Bayesian calibration (rather than approximate Bayesian computation); the window-independent Sharpe–Schoolfield activation energy; and the mechanistic, control-weighted E_a decomposition. Energies are in J mol⁻¹, entropies and heat capacities in J mol⁻¹ K⁻¹, temperatures in kelvin, $R = 8.314$ J mol⁻¹ K⁻¹; the written equations mirror the implementation (`enzyme_cost.py`, `mmrt.py`, `unfolding.py`, `sectors.py`, `sharpe_schoolfield.py`, `ea_dissection.py`).

4.1 The complete model, in four layers

In plain terms, each etc-GEM turns a genome plus enzyme parameters into a predicted growth rate at any temperature, in four stacked layers common to both organisms. The first says growth requires enzymes and the cell has a limited budget of them. The second makes each enzyme’s efficiency temperature-dependent — enzymes speed up with warming, then unfold and fail when hot. The third fixes how the enzyme budget is split among broad functional groups, using measured proteome data, including how that split shifts with growth rate. The fourth is the epistemic stance: the emergent model is a genuine forward prediction — every number grounded in independent data, nothing fit to the growth curve — with any calibration kept as a separate, labelled inverse step. Where the two taxa differ within a layer is flagged as a *departure*, and the departures are collected in Table 4.

Layer 1 — metabolic network with an enzyme budget. We start from a manually curated genome-scale reconstruction (reaction stoichiometry and gene–protein–reaction rules) solved by flux-balance analysis with cobrapy¹⁴, made **enzyme-constrained**: running reaction i at flux v_i requires an amount of enzyme $v_i/k_{cat,i}$, where $k_{cat,i}$ is the turnover number, and the total enzyme mass is capped by a shared **proteome pool**, so the cell must allocate a finite protein budget across reactions (Equation 2, Equation 6). *Departure.* For *E. coli* the base is iML1515¹⁵ as the pre-built GECKO ecModel eciML1515^{9,10}, carrying reference turnovers for 2,560 enzymes; for *M. maripaludis* we build the sMOMENT enzyme layer here on the plain iMR539 reconstruction¹⁶, mapping 527 enzymatic reactions to UniProt masses. A **curation** step differs likewise: *E. coli* required closing four reactions that consume O₂ at negligible enzyme cost — two quinol/menaquinol monooxygenases, a malate oxidase and a cuprous oxidase — which otherwise form a futile O₂ cycle (75% of gross O₂ turnover at the rich-medium optimum) that scrambles the respiratory read-out; closing them routes O₂ through the enzyme-costed terminal oxidase at 0.3% growth cost (catalase and superoxide dismutase, whose high turnover makes their low cost correct, are retained). *M. maripaludis* required a carbon-honest autotroph curation instead: folding the dispensable archaeum glycoprotein out of biomass, repairing a selenocysteinyl-tRNA recycling gap, closing organic-carbon leaks and freeing the pinned methane exchange, giving growth with 99.98% of biomass carbon from CO₂.

Layer 2 — a temperature-dependent enzyme envelope. Each enzyme’s turnover rises with temperature following macromolecular rate theory (MMRT)¹¹ — an Arrhenius law whose activation energy itself changes with temperature (a heat-capacity term), giving a curved, single-peaked response — and above a threshold the

enzyme unfolds, modelled by a two-state native unfolded equilibrium (after⁴), the **native fraction** $f_{N,i}(T)$, which collapses around the enzyme’s melting temperature T_m . The effective per-flux cost combines both effects (Equation 5), rising both when turnover is slow (cold) and when the enzyme denatures (hot); keying the collapse on T_m ties the upper thermal limit to protein *stability*, distinct from the rising-limb activation energy. *Departure*. The provenance of the per-enzyme thermal parameters differs by organism: for *E. coli* the melting temperatures come from the measured melting proteome¹⁷ and the optima from a sequence-based predictor¹⁸ (~90% grounded, rest at dataset means), refined by DLTKcat¹⁹; for *M. maripaludis* the energy-core turnovers are measured specific activities²⁰ converted to k_{cat} , but the optima and melting temperatures are a documented mesophile prior (the sequence/meltome predictors were not run for the archaeon).

Layer 3 — proteome allocation. The enzyme budget is split into three broad measured **proteome sectors**^{12,13} — metabolic (enzymes carrying flux), biosynthesis (dominated by ribosomes, with a translation cap on growth), and maintenance (housekeeping and stress) — with a temperature-dependent non-growth maintenance NGAM(T) that lifts the hot-side ATP burden and rounds the optimum, and an optional growth law that couples the ribosomal fraction to growth rate (Equation 6–Equation 8). *Departure — the scientifically load-bearing one.* The allocation **law** differs between the taxa and is one of the two terms that explain the activation-energy difference: *E. coli* follows the bacterial Scott growth law, in which the ribosomal fraction *rises* with growth rate (slope $s \approx 0.33$), shrinking the metabolic sector at higher temperature and buffering the activation energy down; *M. maripaludis* instead maintains a **constant** ribosomal fraction and full methanogenesis capacity across growth rate⁸ — a documented alternative allocation strategy — so its slope is flat ($s = 0$) and the buffer vanishes. Maintenance is grounded per organism too: an *E. coli* NGAM amplitude for the bacterium, the **measured** *M. maripaludis* maintenance energy²¹ for the methanogen.

Layer 4 — an emergent a-priori prediction, with calibration kept separate. For both organisms the design stance has two distinct parts. *The emergent model is the prediction:* every input is grounded in independent data (the per-enzyme thermal parameters of Layer 2, the measured sector allocation of Layer 3, and the literature magnitude scalars P^{lit} and σ), and nothing is fit to the growth curve, so both the rate magnitude and the whole-organism activation energy **emerge and can be tested** against an independent benchmark. *Calibration is a separate, labelled step:* a Bayesian inverse analysis (Equation 11; the global correction scalars are tabulated in the Supplement) then asks what corrections the measured curve *demand*s of the few borrowed or uncertain global scalars — it does not re-fit the grounded per-enzyme parameters — and the emergent prediction and the tuned posterior are kept distinct, so the a-priori gap is quantified rather than papered over.

4.2 Inputs and data provenance: what the model uses for each enzyme

Each model is **per-enzyme**: every modelled enzyme (2,560 for *E. coli*; 527 for *M. maripaludis*) carries its own inputs, applied individually rather than as one global response shared across the proteome. For each enzyme the model uses four things:

- **Flux** — *not an input but a predicted output* of the flux-balance optimisation.
- **Turnover** $k_{\text{cat}}(T)$ — a reference turnover shaped by an MMRT temperature response. For *E. coli* the reference is the GECKO ecModel value^{9,10}, refined by DLTKcat¹⁹ where a fit exists; for *M. maripaludis* the reference is set by source priority — measured specific activities for the energy-conserving core²⁰, DLTKcat for the biosynthetic bulk (corrected for a systematic archaeal underprediction with cited literature/BRENDA overrides for the highest-cost offenders and a 1 s^{-1} floor), and a dataset-mean fallback.
- **Thermal optimum** T_{opt} and **curvature** $\Delta C_p^\ddagger$ — the optimum is sequence/ML-predicted per enzyme for *E. coli*¹⁸ and a mesophile prior for *M. maripaludis*; the curvature is, for both, a single independent literature MMRT prior ($-4 \text{ kJ mol}^{-1} \text{ K}^{-1}$), deliberately not fitted, so the activation energy emerges.
- **Melting temperature** T_m (setting the native fraction) — from the measured *E. coli* melting proteome¹⁷ for the bacterium and a mesophile prior for the archaeon.

Only the pool budget scalars (P^{lit} , σ) and the sector fractions are whole-cell quantities; everything else is per enzyme. Per-enzyme grounding coverage is high but honest: for *E. coli* ~90% of enzymes have data-grounded optima and melting temperatures (the rest at dataset means); for *M. maripaludis* every enzymatic reaction

maps to a UniProt mass, the energy core (11 reactions) carries measured kinetics, 293 reactions carry DLTKcat predictions and 223 a mean fallback, while the optima/melting temperatures are the mesophile prior throughout — a stated limitation refined by the calibration’s envelope knobs.

4.3 Enzyme abundance and the compensation question

For both models, per-enzyme **abundances are not measured or set individually**: each enzyme’s proteome-mass usage is $\text{flux}/k_{\text{cat}}(T) \times \text{MW}$, drawn from the shared proteome pool that the optimiser allocates across all enzymes, with measured fractions supplied only at the coarse sector level. Per-enzyme abundance is therefore an **optimisation outcome, not an input** — a structural property of the GECKO/sMOMENT pool common to both taxa. This has a consequence worth stating plainly: as temperature rises and an enzyme’s k_{cat} falls, or the enzyme begins to unfold, carrying the same flux costs more enzyme mass, so the optimiser draws more of the pool toward those steps — an implicit, mass-balance “compensation” — until the pool binds and growth is throttled. But this is **least-cost allocation under a fixed budget, not regulatory sensing**: no mechanism detects a falling product and up-regulates a gene. The models therefore capture the *demand* side of thermal compensation (more enzyme needed when catalysis is poor) but not active feedback regulation — a genuine limitation of both.

4.4 Governing equations

Enzyme-constrained flux balance. Growth is the biomass flux v_{bio} found by linear programming under mass balance and enzyme capacity. At steady state

$$\mathbf{S} \mathbf{v} = \mathbf{0}, \quad (1)$$

where $\mathbf{S}$ is the stoichiometric matrix and $\mathbf{v}$ the flux vector ($\text{mmol gDW}^{-1} \text{h}^{-1}$) with lower/upper bounds encoding reversibility and medium availability. Each reaction is limited by the amount of catalytically competent (native) enzyme,

$$0 \leq v_i \leq k_{\text{cat},i}(T) f_{N,i}(T) [E]_i, \quad (2)$$

with turnover $k_{\text{cat},i}(T)$ (Equation 3), total enzyme $[E]_i$, and native fraction $f_{N,i}(T) \in [0, 1]$ (Equation 4) — only folded protein catalyses, while denatured protein still occupies the budget. Rather than an enzyme variable per reaction, we impose the equivalent **sMOMENT** total-protein-pool form: a shared, sector-partitioned budget bounds the summed per-flux enzyme demand (Equation 6).

Temperature-dependent turnover. Each enzyme’s turnover follows transition-state (Eyring) theory with a macromolecular-rate-theory heat-capacity term¹¹,

$$k_{\text{cat},i}(T) = \frac{k_B T}{h} \exp\left(-\frac{\Delta G_i^\ddagger(T)}{RT}\right), \quad \Delta G_i^\ddagger(T) = \Delta H_i^\ddagger + \Delta C_{p,i}^\ddagger(T - T_0) - T(\Delta S_i^\ddagger + \Delta C_{p,i}^\ddagger \ln(T/T_0)), \quad (3)$$

with Boltzmann and Planck constants k_B, h , reference T_0 , and per-enzyme activation enthalpy, entropy and heat capacity $\Delta H_i^\ddagger, \Delta S_i^\ddagger, \Delta C_{p,i}^\ddagger$. A negative $\Delta C_p^\ddagger$ makes $\ln k_{\text{cat}}$ single-peaked, so the **rising-limb activation energy emerges from Equation 3** rather than being imposed. Turnover is carried relative to its value at each enzyme’s optimum $T_{\text{opt},i}$, $\hat{k}_i(T) = k_{\text{cat},i}(T)/k_{\text{cat},i}(T_{\text{opt},i})$, with $\Delta H_i^\ddagger$ fixed by the peak condition $d \ln k_{\text{cat}}/dT = 0$ at $T_{\text{opt},i}$.

Two-state unfolding and the effective cost. The high-temperature collapse is set by denaturation: a native $\leftrightarrow$ unfolded equilibrium keyed on the melting temperature $T_{m,i}$,

$$f_{N,i}(T) = \frac{1}{1 + \exp(-\Delta G_{u,i}(T)/RT)}, \quad \Delta G_{u,i}(T) = \Delta H_{TH,i} + \Delta C_{p,u,i}(T - T_H) - T\Delta S_{TS,i} - T\Delta C_{p,u,i} \ln(T/T_S), \quad (4)$$

with universal convergence temperatures $T_H = 373.5 \text{ K}$, $T_S = 385 \text{ K}$, and unfolding parameters solved from $T_{m,i}$ (with T_{90} or protein length) via $\Delta G_{u,i}(T_{m,i}) = 0$ and $\Delta G_{u,i}(T_{90}) = -RT_{90} \ln 9$. Combining anchored turnover and native fraction, the per-flux enzyme cost is

$$c_i(T) = \frac{\text{base}_i}{\hat{k}_i(T) f_{N,i}(T) \lambda_k}, \quad \text{base}_i = \frac{\text{MW}_i}{k_{\text{cat},i}(T_{\text{opt},i}) \cdot 3600}, \quad (5)$$

with enzyme mass MW_i ($\text{kDa} = \text{g mmol}^{-1}$), the factor 3600 converting s^{-1} to h^{-1} , and a global turnover-level multiplier λ_k (`kcat_scale`, nominal 1). Denatured enzyme still occupies the budget but does not catalyse, so cost is inflated by $1/f_{N,i}$ — the origin of the falling limb, keyed on stability T_m and thus decoupled from the rising-limb E_a .

Proteome sectors, the pool budget, and the growth law. The total modelled protein P_{tot} is split into coarse sectors^{12,13} summing to one — metabolic f_{metab} , biosynthesis/ribosomal f_{bio} , and maintenance f_{maint} — and growth maximises v_{bio} under a metabolic-pool bound and a biosynthesis (translation) cap,

$$\sum_i c_i(T) (v_i^+ + v_i^-) \leq f_{\text{metab}} P_{\text{tot}}, \quad \kappa v_{\text{bio}} \leq f_{\text{bio}} P_{\text{tot}}, \quad (6)$$

where $v_i^\pm \geq 0$ are forward/reverse fluxes and κ (`translation_coeff`) is the ribosomal demand per unit biomass flux, auto-calibrated once at build so the two caps co-limit at the nominal split (enabling sectors does not move nominal growth). The metabolic budget is grounded independently,

$$P_{\text{metab}} = f_{\text{metab}} P_{\text{tot}}, \quad P_{\text{tot}} = \sigma P^{\text{lit}}, \quad (7)$$

with in-vivo saturation σ and literature total protein P^{lit} . Optionally the sectors follow a proteome-conserving growth law in which the ribosomal and metabolic fractions move with growth rate at a common slope s , $f_{\text{bio}}(\mu) = f_{\text{bio},0} + s\mu$ and $f_{\text{metab}}(\mu) = f_{\text{metab},0} - s\mu$ ($\mu = v_{\text{bio}}$), keeping both caps linear,

$$(\kappa - sP_{\text{tot}}) v_{\text{bio}} \leq f_{\text{bio},0} P_{\text{tot}}, \quad \sum_i c_i(T) (v_i^+ + v_i^-) + sP_{\text{tot}} v_{\text{bio}} \leq f_{\text{metab},0} P_{\text{tot}}. \quad (8)$$

The slope s is the scientifically load-bearing organism-specific quantity: it is set from each organism's own proteome data (below), and its value is the origin of the allocation term in the E_a decomposition.

Maintenance. A temperature-dependent non-growth maintenance ATP demand rises with temperature²²,

$$\text{NGAM}(T) = a_m \left(1 - b \exp\left[\frac{-E_m}{k_B} \left(\frac{1}{T_{\text{ref}}} - \frac{1}{T} \right) \right] \right), \quad \text{floored at } T_{\text{ref}}, \quad (9)$$

with amplitude a_m , $b \approx 0.62$, apparent activation energy $E_m \approx 0.5$ eV, $T_{\text{ref}} = 298.15$ K and $k_B = 8.617 \times 10^{-5}$ eV K⁻¹, imposed as $v_{\text{ATPM}} \geq \text{NGAM}(T) f_{\text{maint}}/f_{\text{maint}}^{\text{nom}}$; the growth-associated maintenance (GAM) is the ATP demand carried in the biomass reaction. At each grid temperature FBA maximises $\mu(T) = \max_{\mathbf{v}} v_{\text{bio}}$ subject to Equation 1, Equation 6 and the maintenance floor, tracing the organismal TPC.

The activation energy. From $\mu(T)$ we read standard descriptors — the optimum $T_{\text{opt}} = \arg \max_T \mu$, the maximum rate r_{max} , the critical limits $CT_{\text{min}}/CT_{\text{max}}$ (where μ falls to 5% of r_{max}), and the niche width. The naive Boltzmann–Arrhenius activation energy — the slope of $\ln \mu$ against $-1/(k_B T)$ over a rising-limb window — is window-dependent (Results, Section 2.2), so we report instead the **window-independent Sharpe–Schoolfield** E^7 , the rising-limb activation energy of the high-temperature-inactivation model

$$\mu(T) = r_{T_{\text{ref}}} \frac{\exp[(E/k_B)(1/T_{\text{ref}} - 1/T)]}{1 + \exp[(E_h/k_B)(1/T_h - 1/T)]}, \quad (10)$$

fit to the whole calibrated curve by nonlinear least squares at a fixed $T_{\text{ref}} = 20$ °C, where E is the reported E_a and E_h, T_h describe the high- T deactivation. The same fit is applied identically to model and observed curves and to both organisms.

Bayesian calibration. Each model is calibrated to one measured growth TPC with a gradient-free ensemble sampler (emcee), warm-started at the posterior mode and run to an autocorrelation-based stop. The deterministic simulator $\theta \mapsto \mu(T)$ is combined with a Gaussian discrepancy likelihood

$$\ln L(\theta) = -\frac{1}{2} \sum_t \left[\frac{(\mu_t^{\text{obs}} - \mu(T_t; \theta))^2}{\text{sd}_t^2 + \sigma_{\text{disc}}^2} + \ln(2\pi(\text{sd}_t^2 + \sigma_{\text{disc}}^2)) \right], \quad (11)$$

with per-point measurement SD sd_t (zero where none is reported) and a free model-discrepancy scale σ_{disc} . Only a small set of **global scalars** is freed — $\{\lambda_k$ (`kcat_scale`), σ , dT_{opt} , `topt_scale`, `dCp_scale`, dT_m ,

tm_scale , f_{metab} , f_{maint} , ngam_scale , ngam_steepness , σ_{disc} — each transforming the *whole* grounded per-enzyme distribution while preserving its shape (a uniform shift, stretch or rescale), never a per-enzyme free parameter. Positive-only parameters are sampled in log space; priors are provenance-based (measured sector fractions tight; the in-vivo saturation a broad Normal on 0.45; magnitude and envelope levers LogNormal about their nominal). The emergent (nothing-fit) model and the calibrated posterior are kept distinct throughout.

The control-weighted activation-energy decomposition. To ask what sets each E_a we decompose the Sharpe–Schoolfield E , by metabolic control analysis, into a control-weighted mean of the per-enzyme rising-limb activation energies and four named couplings relative to the naive (unweighted) enzyme mean:

$$E_a^{\text{org}} = \overline{E_i} + \underbrace{\left(\sum_i \hat{C}_i E_i - \overline{E_i} \right)}_{\text{control weighting}} + \Delta_{\text{alloc}} + \Delta_{\text{maint}} + \Delta_{\text{agg}}. \quad (12)$$

The construction is non-circular: the per-enzyme E_i — the rising-limb Arrhenius activation energy of each enzyme’s turnover $k_{\text{cat},i}(T)$ alone (the folding term is the high- T deactivation, contributing a negligible -0.008 eV control-weighted) — are model *inputs*, and the result is how the organism value aggregates and departs from them. The growth flux-control coefficients $\hat{C}_i = \partial \ln \mu / \partial \ln k_{\text{cat},i}$ are computed by symmetric perturbation and re-solve and normalised over the controlling set. The **allocation** term is isolated by disabling the growth-law coupling (Equation 8), the **maintenance** term by flattening NGAM(T), and the **aggregation** term is computed directly as the Sharpe–Schoolfield E of the kinetic-backbone aggregate curve minus the control-weighted per-enzyme mean. Both organisms are analysed on the calibrated model at their operating point, each carrying its own sector/growth-law layer, with the uncostered energy side-reactions closed. For Figure 1 the saved calibration draws were propagated through Equation 10 to obtain the E_a posteriors.

4.5 Model hierarchy: how the components link

The model is a hierarchy that turns grounded per-enzyme thermal parameters into an organismal curve, each level feeding only the next, identically for both taxa. **(i)** Every enzyme carries three grounded thermal parameters — its optimum $T_{\text{opt},i}$, its melting temperature $T_{m,i}$ and its turnover curvature $\Delta C_{p,i}^\ddagger$. **(ii)** These set the two per-enzyme temperature responses: the optimum-anchored turnover $\hat{k}_i(T)$ (Equation 3) and the native fraction $f_{N,i}(T)$ (Equation 4). **(iii)** The two combine into an effective per-flux enzyme cost $c_i(T)$ (Equation 5). **(iv)** The costs enter the sector-partitioned proteome constraints — metabolic pool, biosynthesis cap and temperature-dependent maintenance (Equation 6, Equation 9) — optionally coupled through the growth law (Equation 8). **(v)** FBA maximises biomass flux at each temperature, tracing the TPC $\mu(T)$. **(vi)** The curve is reduced to descriptors and the Sharpe–Schoolfield activation energy (Equation 10). **(vii)** For calibration and the decomposition, the global correction knobs (Supplement) deform the whole per-enzyme distribution at level (i). The forward chain is thus *grounded parameters* $\rightarrow$ *per-enzyme responses* $\rightarrow$ *cost* $\rightarrow$ *sector constraints + growth law + NGAM(T)* $\rightarrow$ *FBA growth* $\rightarrow$ *descriptors*, with no growth-curve information entering anywhere in the emergent model.

4.6 Relationship to the Li et al. 2021 yeast etc-GEM

Both models are direct descendants of the yeast etc-GEM of⁴, and we **follow it for the thermal core**: the transition-state turnover $k_{\text{cat}}(T)$ (Equation 3), the two-state native-fraction denaturation keyed on T_m (Equation 4), a temperature-dependent maintenance term, and the enzyme-constrained proteome-pool idea. We **depart from and upgrade** it in five deliberate ways. First, **organism**: rather than *S. cerevisiae*, we build a bacterium and an archaeon — to our knowledge the first etc-GEMs of their kind for either taxon — with the archaeon requiring the sMOMENT enzyme layer built here rather than a pre-existing GECKO ecModel. Second, **grounded not fitted per-enzyme parameters**: we fix each enzyme’s T_{opt} , T_m and curvature from independent data, so the TPC is an a-priori forward prediction, whereas Li et al. infer the per-enzyme thermal parameters by fitting the growth data — the opposite epistemic stance, and grounding is what lets us *test* the emergent activation energy against an independent benchmark. Third, a **sector-partitioned proteome** with an explicit translation cap, replacing the single undivided pool, which makes allocation an attributable control on the rate. Fourth, an **organism-specific allocation law** — the Scott rising-ribosome law for *E. coli*, the Müller constant-ribosome strategy for *M. maripaludis* — rather than a single fixed coupling.

Fifth, **inference by likelihood, not ABC**: we calibrate a low-dimensional set of global correction knobs by likelihood-based ensemble MCMC (emcee), kept as a separate inverse layer on top of the emergent prior, exploiting the fact that the etc-GEM is a deterministic simulator with a Gaussian residual model; keeping the emergent prediction and the tuned posterior separate is what lets us report an honest a-priori gap. For quantifying the activation energy we further add the window-independent Sharpe–Schoolfield E (Equation 10) and the control-weighted E_a decomposition (Equation 12), neither of which appears in the yeast model.

4.7 Validation

For each organism the emergent (nothing-fit) model is first tested a-priori against an independent exact-strain growth TPC, and only then calibrated. *E. coli* is validated against the K-12 MG1655 rich-medium (BHI) curve of⁵ over 7–46 °C; *M. maripaludis* against the type-strain H /CO curve of⁶ over 15–52 °C. The reported activation energy is the window-independent Sharpe–Schoolfield E of Equation 10, applied identically to model and observed curves; the test is whether the *modelled* E matches the *observed* E , and whether it lands near the “universal” ~ 0.65 eV benchmark for the respiring bacterium. It does for both: after calibration the *E. coli* model gives $E_a = 0.68$ eV against an observed 0.56 (on the benchmark), and *M. maripaludis* gives 1.06 against an observed 1.04, with the two calibration posteriors cleanly separated (Results, Section 2.1, Section 2.2). Because the per-enzyme activation energies are model inputs, this agreement on the organism-level value is a genuine prediction rather than a recovered fit.

4.8 Taxon-specific construction and departures

The two models instantiate the same framework but differ, layer by layer, in ways that are either technical (base reconstruction, parameter provenance) or scientifically load-bearing (the allocation law). Table 4 summarises the contrasts; the two most consequential are the kcat provenance for the energy core and — the term that carries part of the E_a difference — the allocation law.

Base network and enzyme layer. *E. coli* starts from iML1515¹⁵ as a pre-built GECKO ecModel (eciML1515)^{9,10} carrying reference turnovers and masses for 2,560 enzymes; *M. maripaludis* starts from the plain iMR539 reconstruction¹⁶, onto which we build the sMOMENT enzyme layer here (527 enzymatic reactions mapped to UniProt masses).

Curation. *E. coli* required closing four uncostered O -sink reactions (two quinol monooxygenases, a malate:O oxidoreductase, a cuprous oxidase) that form a futile O cycle. *M. maripaludis* required a carbon-honest autotroph curation: folding the dispensable archaeum glycoprotein out of biomass, repairing a selenocysteinyI-tRNA recycling gap, closing organic- carbon leaks and freeing the pinned methane exchange, and supplying two non-synthesisable vitamin cofactors as trace supplements.

Turnover provenance. *E. coli* uses the GECKO built-in k_{cat} refined by DLTKcat¹⁹. For *M. maripaludis* the energy-conserving core was set from measured specific activities (the electron-bifurcation hydrogenase/heterodisulfide-reductase complexes²⁰, converted to k_{cat} via complex masses), the biosynthetic bulk from DLTKcat corrected for a systematic archaeal underprediction (28% of predictions below 1 s⁻¹) with cited literature/BRENDA overrides for the highest-cost offenders and a documented 1 s⁻¹ floor, and a mean fallback; Mcr — the E_a target — was sourced explicitly ($k_{\text{cat}} \approx 59$ s⁻¹, 3–294 s⁻¹ range, swept in Section 2.4).

Optima and stability. *E. coli* $T_{\text{opt},i}$ and $T_{m,i}$ are grounded per enzyme (sequence-based optimum predictor¹⁸; measured melting proteome¹⁷) for $\sim 90\%$ of enzymes. For *M. maripaludis* these predictors were not run; both use a documented mesophile prior (a flagged limitation refined by M4’s dT_{opt}/dT_m).

The allocation law (load-bearing departure). *E. coli* uses the bacterial Scott growth law¹³: the ribosomal fraction *rises* with growth rate (slope $s \approx 0.33$), which shrinks the metabolic sector at higher temperature and buffers E_a downward. *M. maripaludis* holds a **constant** ribosomal fraction across growth rate⁸ — a documented alternative allocation strategy — so its slope $s = 0$ (a flat law), and the allocation buffer vanishes. This physiological difference, not a modelling choice, is one of the two terms that explain the E_a difference (Results, Section 2.3).

Maintenance, medium and saturation. Maintenance NGAM(T) uses an *E. coli* amplitude for the

bacterium and the **measured** *M. maripaludis* maintenance energy²¹ for the methanogen. Validation/calibration is against BHI-rich growth (Van Derlinden 2012⁵) for *E. coli* and H /CO growth (Jones 1983⁶) for the methanogen. Both take the in-vivo saturation σ from the same literature range^{23,24}. Full per-model construction, parameter and provenance tables, and robustness analyses are in the Supplement.

Table 4: Where the two etc-GEMs depart, by model layer. The allocation-law row (Scott rising vs Müller constant ribosome fraction) is the scientifically load-bearing difference.

Layer	<i>E. coli</i> (respiration)	<i>M. maripaludis</i> (methanogenesis)
Base GEM	iML1515 ¹⁵ → GECKO eciML1515	iMR539 ¹⁶ → our sMOMENT layer
Curation	4 uncosted O sinks closed	carbon-honest autotroph (archaellin/SeCys/leaks)
k_{cat}	ecModel built-in + DLTKcat	measured core ²⁰ + DLTKcat + overrides + floor
T_{opt}/T_m	sequence/meltome-grounded (~90%)	mesophile prior
Allocation law	Scott: ribosome fraction rises with μ ($s \approx 0.33$)	Müller: constant ribosome fraction ($s = 0$)
Maintenance	<i>E. coli</i> NGAM(T)	measured NGAM ²¹
Medium / curve	BHI rich, Van Derlinden 2012	H /CO , Jones 1983
Saturation σ	0.45 (0.4–0.5) ^{23,24}	0.45 (0.4–0.5) ^{23,24}

5 Data and code availability

All figures and tables are generated from the analysis outputs of the two etc-GEMs and their cross-organism comparison. The complete model construction for both organisms, the parameter and data-source tables, the window-sensitivity audit, the calibration diagnostics, and the robustness analyses are provided in the Supplement.

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